The Cost of Patient Autonomy in Cross-Silo Federated Learning

Ph.D. Thesis Proposal Defense

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Roadmap of this talk

What I am asking for today

Approval of a **measurement** thesis: one axis is already executed and closed; the proposal is to build the second axis and join them.

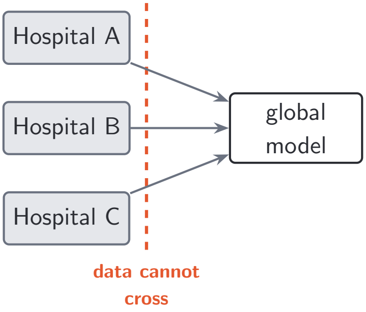
Green marks results already in hand.

Gray marks proposed work.

The Problem

Hospitals cannot pool patient data — but need to

- Clinical prediction models need **scale and diversity** to generalize across institutions.
- Patient records cannot leave the hospital: HIPAA, GDPR, institutional liability, and genuine re-identification risk.
- A single hospital's model **overfits its own case mix** — it does not transfer.

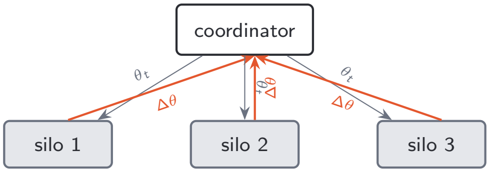


Measured in this work

Training on one silo alone (no federation) drops AUROC to **0.616** on average, and as low as **0.516** — barely better than a coin flip — versus **0.666** pooled.

Federated learning is the standard answer

- Each hospital trains **locally**; only model weights are shared.
- A coordinator averages the weights (**FedAvg**, McMahan 2017) and broadcasts the new global model.
- Raw records never move. Repeat for R rounds.



FedAvg's well-known weakness — sensitivity to non-IID data — turns out to be exactly what makes this thesis measurable. More on that later.

But federation alone does not give patients control

- FL keeps data *inside* the hospital. It says nothing about whether **the patient** agreed to be in the model.
- Consent today is a paper form: coarse, one-time, effectively irrevocable once training has happened.
- GDPR Art. 17 ("right to erasure") and emerging U.S. rules point the other way: consent should be **fine-grained, dynamic, revocable**.

The literature's move

A growing body of work adds a **blockchain consent layer**: smart contracts record grant/revoke, so a patient can withdraw at any time and the next training round must respect it.

The gap: everyone builds it, nobody prices it

The consent-on-blockchain literature is **overwhelmingly architectural**:

- "We designed a system where patients can revoke consent."
- "Here is the contract, here is the workflow."
- Evaluation stops at *feasibility* — it compiles, it runs, gas is reported for a toy workload.

Patient autonomy is treated as an **unqualified good**: a feature to be added, never a cost to be measured.

What is missing

Nobody has asked:

1. What does consent governance **cost per training round**?
2. What does revocation do to **model accuracy**?
3. Where do those two costs make the system **stop working**?

Why the question is not rhetorical

Revocable consent breaks an assumption every FL algorithm makes.

Standard FL assumes

A **fixed** client population and a **stationary** data distribution across rounds. Convergence guarantees are stated in those terms.

Consent-gated FL delivers

A population that **changes every round**, in a direction that is *not* random — patients who withdraw are correlated with their diagnosis, their hospital, their outcome.

- Every revocation is also an **on-chain transaction** — gas, latency, and a consent-set resolution the coordinator must perform before it can start the round.
- So autonomy is paid for **twice**: once in systems overhead, once in model utility.

The reframe

Patient autonomy is not a feature to be built.

It is a **cost to be measured**.

The field's framing

"Can we give patients revocable consent in federated healthcare ML?"

Answer: yes, obviously — and it has been answered many times.

This thesis's framing

"**What does it cost**, and **where does it break**?"

Answer: a two-axis measurement and a frontier with explicit breaking points.

The two axes of cost

Axis 1 — Systems cost [PLANNED]

On-chain governance overhead added to *every* round:

- consent-set resolution
- grant/revoke transactions (gas)
- encrypted-record retrieval
- latency, bandwidth

Axis 2 — Utility cost [EXECUTED]

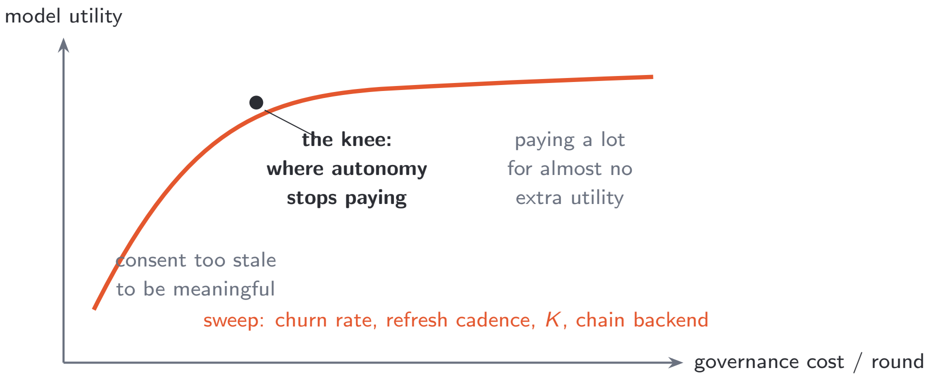
Accuracy lost because revocable consent makes the training population **non-stationary**:

- who withdraws, and when
- whether withdrawal shifts the distribution
- whether a whole institution exits

The deliverable

Not two separate numbers — a **joint cost–utility frontier**, swept over the same consent knobs, with the breaking points marked on it.

What a frontier buys you that a feature list does not



A frontier tells a hospital consortium *which operating point to pick*. “We support revocation” does not.

(H3 — shape is hypothesized, not yet measured.)

Thesis Statement and Research Questions

Thesis statement

Thesis

This dissertation **quantifies the two-axis cost of patient autonomy** — governance overhead and model-utility degradation — in a consent-gated cross-silo federated healthcare system, and identifies the configurations and **breaking points** at which patient-controlled federated clinical prediction remains practical.

This is

A **measurement and systems** contribution. The artifacts are cost–utility curves and the breaking points on them.

This is not

A new FL algorithm. A peak-accuracy result. A blockchain system paper — that component is already published.

A standard I hold myself to

The evaluation must surface a **non-obvious** finding.

Would not count

"More dropout lowers accuracy."

True, trivial, and predictable without running anything. A thesis that only establishes this has measured nothing.

Would count — and does

"**Who** leaves matters more than **how many** leave."

Overturns the headcount intuition, and changes what a consortium should actually worry about. **[EXECUTED]**

Each research question below carries an explicit non-obvious-finding target, not just a hypothesis.

RQ1 — The systems-cost axis [PLANNED]

Question

How does on-chain consent-governance overhead scale with **churn rate**, **consent-refresh cadence**, **federation size** K, and **chain backend** (local / permissioned / public)?

H1

Per-round governance cost is dominated by **consent-set resolution** and grows with churn rate, crossing a **backend-dependent threshold** beyond which per-round enforcement is infeasible and consent must be **amortized** across rounds.

Non-obvious target

Locate that threshold, and characterize the **amortization–staleness tradeoff** it forces: amortizing consent means training on consent state that is knowingly out of date.

RQ2 — The utility-cost axis [EXECUTED]

Question

How does model utility degrade as a function of consent dynamics, and do **physically distinct churn regimes** degrade it differently?

H2

The regimes are **not interchangeable**: transient churn is near-harmless because contributions re-enter, whereas whole-silo departure inflicts a non-IID shock that **dominates all per-patient effects**.

Non-obvious target — confirmed

Show that *who* leaves matters more than *how many* leave, overturning the headcount model. Confirmed by a count-matched isolation test over five seeds, and shown **model-independent** by a full rerun with a higher-capacity client.

Question

Where is the joint cost–utility frontier, and what configurations keep the system in a **practical operating region**?

H3

There exist consent-refresh cadences that recover **most of the utility** of per-round enforcement at **a fraction of the governance cost** — i.e. the frontier is sharply **knee-shaped**, not linear.

Non-obvious target

Identify the knee and the deployment regimes it implies. If the frontier is knee-shaped, "refresh consent every round" — which every system paper assumes — is the *wrong* default.

RQ4 — Multi-project consent [PLANNED]

Question

Can a **single consent layer** govern patients enrolled in several federated studies at once (a patient $\times$ study matrix), and what is the **marginal governance cost** per additional study?

H4

Multi-project consent adds **bounded, sub-linear** on-chain overhead, because consent state is **shared infrastructure** rather than per-study duplication.

Non-obvious target

Demonstrate the framework generalizes beyond a single disease model **without building a second one** — the marginal cost, not a second end-to-end system, is the evidence.

Contributions

C1 **A dual-axis measurement of the cost of patient autonomy.** First joint characterization of governance overhead and utility degradation under revocable consent, as cost–utility curves with explicit breaking points. (RQ1, RQ3)

C2 **A three-regime model of consent churn.** Separates transient departure, permanent withdrawal, and whole-silo exit into physically distinct regimes; confirms *who* $>$ *how many* via a count-matched isolation test; shows it is model-independent. (RQ2) **[EXECUTED]**

C3 **A multi-project consent framework.** A patient $\times$ study consent matrix at the contract level, with measured marginal cost per study. (RQ4)

C4 **Validated, reusable components and methodology.** A scalable multimodal classifier; a patient-controlled blockchain/IPFS consent layer with a **validated on-chain cost-measurement methodology**; a multi-institutional regime-shift analysis. (feasibility) **[EXECUTED]**

C5 **An open, reproducible testbed and benchmark.** Testbed, contracts, churn-schedule generators, and Synthea-derived populations, released open-source. (all RQs)

Scope: what is deliberately in and out

In scope

- Simulated cross-silo consortium, $K{=}3$ and small K
- 30-day readmission on Diabetes 130 (real, 102k encounters)
- Class-weighted FedAvg as the reference instrument
- Consent **governance** and **data autonomy**
- Gas/latency across local, permissioned, public backends

Out of scope

- Live multi-hospital deployment
- Daily EHR transaction processing, billing workflows
- Supply-chain forecasting as a co-equal target
- **Formal privacy guarantees** in Year 1 — see next slide
- Peak accuracy on the clinical task

One honesty commitment I want on the record

The Year-1 claim is consent governance and data autonomy — not privacy

FedAvg alone provides **leaky data-locality**, not a differential-privacy guarantee. Gradients leak. Calling this system "privacy-preserving" would be overclaiming, and I do not.

- Differential privacy (DP-SGD) and secure aggregation are **composable layers** added in a later thrust.
- That thrust *earns* the word "privacy" — and adds a **third axis** (noise $\rightarrow$ accuracy loss) to the same frontier.
- Deferring it is a scoping decision, not an oversight: the two axes must be measured cleanly before a third is layered on.

Foundations: Three Published Components

The proposal does not start from zero

Three published papers supply the instruments this thesis needs.

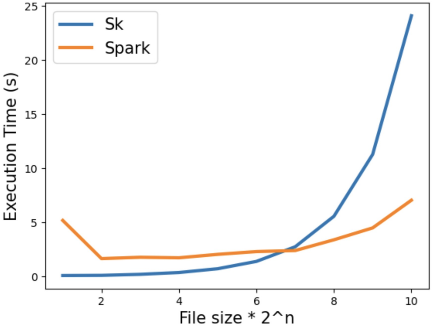
Published work	What it built	What it gives the thesis
Spark benchmarking (cervical cancer)	Scalable Spark/SparkML pipeline vs. scikit-learn and a CNN	The **compute-placement rule**, and the Year-3 image baseline
Patient-centric blockchain EHR	Ethereum + IPFS consent layer with grant/revoke	The **consent layer** and a **validated on-chain cost methodology** — the RQ1 instrument
Supply-chain regime shift (COVID)	Lead-time models across a multi-institutional supply chain	Documented **cross-institution regime shift** — the real-world cousin of non-IID shock

These are *feasibility and instrumentation*, not the thesis contribution itself.

Component 1 — Spark benchmarking (published)

What was built. A scalable cervical-cancer detection pipeline on Apache Spark/SparkML, benchmarked against scikit-learn and a CNN across **two modalities**.

- **Tabular** (cervical risk factors): Spark Random Forest reached **0.96** accuracy (0.99 precision, 0.97 recall) after ADASYN balancing of a strongly imbalanced cohort.
- **Scalability**: scaling data 0.27 $\rightarrow$ 217 MB, Spark crossed over to $\approx$ **3.3**$\times$ **faster** than scikit-learn as volume grew.
- **Images** (SipakMed Pap smears): a CNN reached **92.75%** where Spark's general classifiers plateaued near **73%**.



Spark vs. scikit-learn as data volume grows.

Component 1 — what it gives the thesis

The compute-placement rule (empirically grounded)

Distributed / Spark compute **earns its place on large image data** but is **overkill on small tabular data**.

- This rule is **honored, not ignored**, in the thesis plan: Year 1 uses **plain tabular federated training**; Spark and images are deferred to Year 3.
- Without it, the obvious — and wrong — move would be to put the federated readmission task on Spark because it sounds more impressive.
- It also supplies the **Year-3 image task and baseline**, so the image extension does not start cold.

A negative-space contribution: prior work telling me what *not* to build.

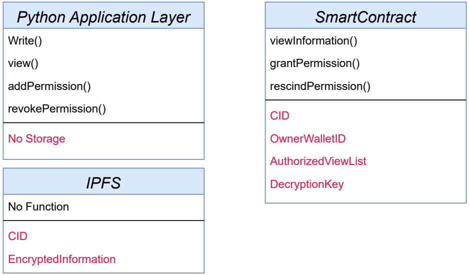
Component 2 — Patient-centric blockchain EHR (published)

A Python application mediates an **Ethereum smart contract** and **IPFS** storage.

- Records encrypted (Fernet), stored **off-chain on IPFS**.
- The contract holds only: content identifier, owner wallet, authorized-viewer list, key.
- Exposes `grantPermission`, `rescindPermission`, `viewInformation`, `isViewerAuthorized`.

Design invariant

No clinical data on the ledger. An immutable, replicated chain holding PHI would be a privacy liability, not a feature.



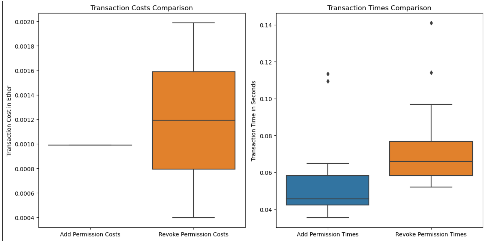
System component overview.

Component 2 — measured cost on a local chain

Ganache (single-node local chain):

Operation	Time (s)	Cost (ETH)
Add permission	0.0521	0.00099
Revoke permission	0.0705	0.00119

- All operations under **0.1 s**; IPFS retrieval typically under **0.3 s**.
- **Revoke costs more than add** — consistently, on both backends.
- Ganache variance is near zero — but it is *one node*, so this is a best case, not a deployment estimate.

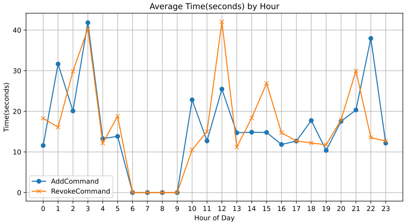


Ganache transaction cost/time distributions.

Component 2 — and on a public chain, it falls apart

Sepolia (public testnet), measured across the day:

- Fastest: add **8.84 s**, revoke **12.01 s**.
- Peaks: revoke **107.99 s** at 1 AM; add **55.09 s** at 12 PM.
- Congestion 6–9 AM CT caused outright **transaction failures** under our wallet settings.
- Revoke cost peaked at **0.0198 ETH** — roughly 20× the Ganache figure.



Sepolia transaction time through the day.

Why this matters for RQ1

Latency is **heavy-tailed**, not Gaussian. A mean would hide the failures — so RQ1 reports **distributions**.

Component 2 — what it gives the thesis

The RQ1 instrument

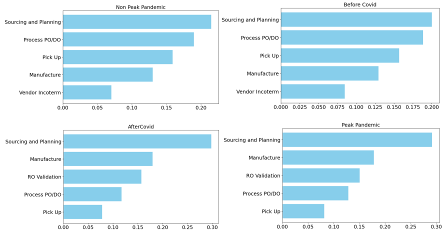
A **validated methodology for measuring on-chain governance cost** — latency and gas per operation, across backends, reported as distributions.

- The consent-layer **design and grant/revoke semantics** carry over directly.
- The 4-orders-of-magnitude spread between Ganache (0.05 s) and congested Sepolia (108 s) is **precisely the H1 threshold question**: at what churn rate does per-round consent enforcement stop keeping up?
- The published paper's own stated future work — "extensive load testing is necessary" — is **exactly what RQ1 does**.

Component 3 — Supply-chain regime shift (published)

Lead-time modeling across a large multi-institutional healthcare supply chain.

- Random Forest regression, $R^2 \geq 0.87$.
- **Feature importance shifts** between normal operation and crisis (pandemic) operation — the same model, the same institutions, different drivers.



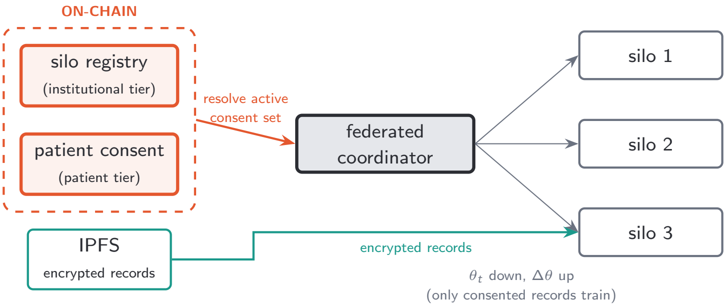
Normal vs. stressed operating regimes.

What it gives the thesis

A **documented, real-world multi-institution regime shift** — the empirical cousin of the non-IID shock studied in RQ2. Cross-institution distribution shift is not a simulation artifact.

System Design

Consent-gated cross-silo FL: the architecture



Before each round — or each **refresh interval**, a swept parameter — the coordinator resolves two things from the ledger: admitted **silos** (mostly static) and active **patients** (dynamic, revocable).

The two-tier consent model

Institutional tier

A **silo registry** contract: which hospitals are admitted to the consortium.

- Mostly **static**.
- Changes are rare but **catastrophic** — a whole distribution slice appears or vanishes.

Patient tier

A **patient-consent** contract: which patients within each silo currently consent.

- Highly **dynamic** and revocable.
- Changes are frequent but individually **small**.

This split is the whole thesis in miniature

The systems cost is driven mostly by the **patient tier** (transaction volume). The utility cost, as the results will show, is driven mostly by the **institutional tier** (distribution shift). *They do not move together* — which is exactly why a joint

Data: one powered primary, one synthetic stressor

Primary: UCI Diabetes 130-US-hospitals

- **101,766** real inpatient encounters, 130 U.S. hospitals, 1999–2008
- Task: **30-day readmission** (binary)
- **11.2%** positive (≈11k positive cases)
- **171** features after preprocessing

Carries the **entire utility axis**.

Stressor: Synthea (synthetic)

- FHIR-native populations at **hospital scale**
- Confined to the **systems axis only**
- There, signal quality is **irrelevant by construction** — only transaction volume and payload size matter

The cervical dataset (858 records, 54 positives) is retained only as a small **signal-validity anchor** — see next slide for why it cannot carry the churn study.

Why the dataset choice was re-made (and it mattered)

What was dropped, and why

Cervical (858 rows) — too small. The binding number is **54 positive cases**. Across 3 silos at up to 70% churn, positives fall to **single digits per silo**: statistically dead for measuring degradation deltas.

Synthea-for-accuracy — wrong tool. Its signal is weak *by construction*, so degradation curves computed on it would measure **generator artifacts**, not utility loss.

Why Diabetes 130 replaces both

It is **real** (so degradation measures genuine clinical signal) *and* **large** (so per-silo churn deltas stay statistically powered).

It collapses an earlier two-source workaround — in which synthetic data would have carried the powered curves — into **one better source**.

The methodology itself is **disease-agnostic**; readmission is chosen for statistical power, not clinical preference.

Preprocessing (identical across every experiment)

- **Target:** `readmitted == "<30"` → 1, else 0.
- **ICD-9 grouping:** `diag_1/2/3` mapped to clinical categories (Circulatory, Respiratory, Digestive, Diabetes, Injury, Musculoskeletal, Genitourinary, Neoplasms, Other).
- **Dropped:** `weight`, `payer_code`, `examide`, `citoglipton` (near-empty or constant).
- **Rare-level collapsing:** categories under 1% frequency folded into `Other`.
- **Scaling:** numeric features standardized using **train-split statistics only**.

Why this is stated explicitly

Every number in this talk — centralized, federated, and every churn delta — comes through **this same pipeline**.

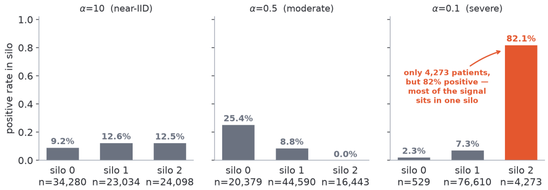
A degradation study is only meaningful if the preprocessing is held **exactly** constant; otherwise the deltas measure pipeline drift.

80/20 stratified train/test split; the test set is **global and never churned**.

Creating heterogeneity: Dirichlet label partitioning

Silos are formed by a **Dirichlet**(α) **label-distribution split**:

- $\alpha \to \infty$: IID, every silo looks the same.
- $\alpha = 0.5$: moderate skew.
- $\alpha = 0.1$: **severe** skew.



This is what makes "who leaves" meaningful

At α=0.1, one silo holds **4,273 patients at 82% positive** while another holds **76,610 at 7.3%**. Losing the small one is nothing like losing the big one.

Honest framing before any results

Peak accuracy is explicitly not the target

The centralized ceiling on this task is ≈ 0.677 AUROC. That is **not the object of study** and chasing it would optimize the wrong quantity.

- What is under study is **how that accuracy moves**, and what governance costs, as patients exercise revocable consent.
- The task's **low signal ceiling is convenient**: it guarantees measured degradation reflects **population change**, not under-fitting headroom.
- The scientific weight rests entirely on the **deltas**.

I will return to this when discussing why nothing amplifies under a stronger model — it is the same fact, seen from the other side.

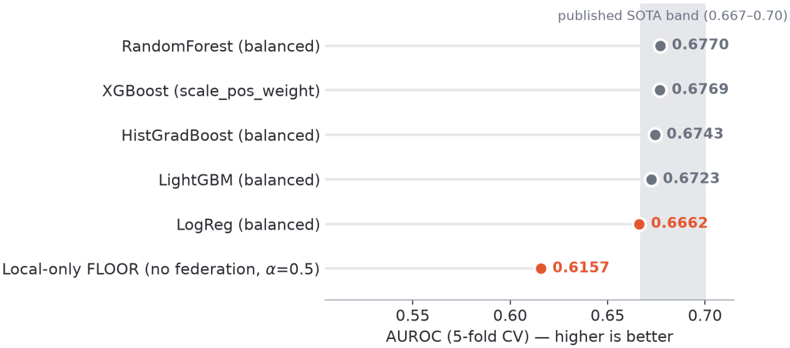
Executed Results: The Utility Axis (RQ2)

What is already executed

Study	Scale	Status
Centralized multi-model benchmark	5 models, 5-fold CV	[EXECUTED]
Local-only floor (no federation)	per-silo, α=0.5	[EXECUTED]
Five-method federated comparison	5 methods $\times$ 3 skew levels	[EXECUTED]
Three-regime consent-churn study	4 rates $\times$ 5 seeds	[EXECUTED]
Whole-silo exit + count-matched control	2 skews $\times$ 5 seeds	[EXECUTED]
MLP amplification check (full grid re-run)	**200 training runs**	[EXECUTED]
Systems-cost instrumentation (RQ1)	3 backends	[PLANNED]
Joint frontier (RQ3), multi-project (RQ4)	—	[PLANNED]

Every figure in this section is regenerated directly from the saved result files.

Anchor 1 — the centralized ceiling and the local floor



- Best centralized: **0.6770** (Random Forest), inside the published **0.667–0.70** band for this task. Model choice barely moves AUROC — the signal ceiling is genuinely low.
- Local-only floor: **0.6157** mean (min **0.5159**). This is the cost of *not* federating.

Anchor 2 — the federated ceiling is lower, and that is a result

Trees are not weight-averageable

Random Forest and XGBoost — the two best centralized models — **cannot be averaged** across silos. FedAvg needs a parameter vector, not a forest.

So the **federated ceiling** is set by the best *averageable* model:

- Logistic regression: **0.6662**
- One-hidden-layer MLP: **0.6700**
- Tree ceiling: **0.6770**

The first measured cost

The **0.007–0.011 AUROC gap** between the tree ceiling and the averageable ceiling is the first concrete, measured cost of federating this task.

It is paid **before any patient withdraws consent at all**.

This is rarely stated in FL papers, which typically compare federated-vs-centralized *within* one model family and so never see it.

Anchor 3 — does the aggregation method matter?

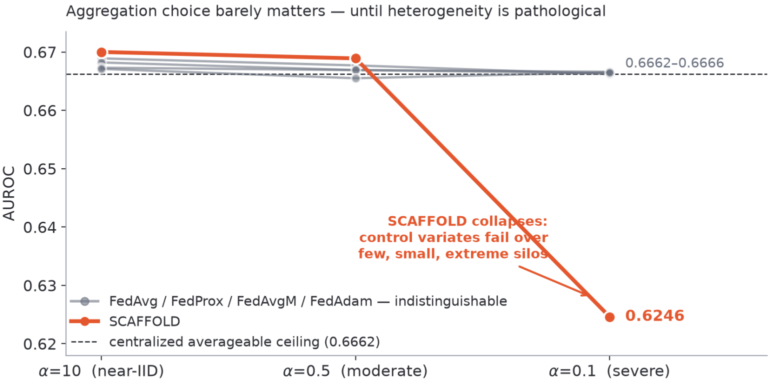
Five server-side aggregation strategies, same client, same data, same splits:

Method	**Mechanism**	**Reference**
FedAvg	weighted parameter average	McMahan 2017
FedProx	FedAvg + client proximal term μ	Li 2020
FedAvgM	server momentum on aggregated delta	Hsu 2019
FedAdam	adaptive server optimizer (FedOpt)	Reddi 2021
SCAFFOLD	client control variates correct drift	Karimireddy 2020

Implementation note that makes this fair

All five operate on **one flat parameter vector**. Deltas, proximal terms, and control variates are therefore **model-agnostic** — swapping the client model does not change the aggregation code at all. This matters enormously later.

Anchor 3 — the answer: barely, until it is pathological



- Four methods are **statistically indistinguishable** at every skew level (0.6655–0.6700).
- **SCAFFOLD collapses at** $\alpha=0.1$: **0.6246**, a documented small-K failure mode — control variates are estimated from few, small, extreme silos and inject noise instead of correcting drift.

Why FedAvg is the empirically justified instrument

The weak argument (not used)

"FedAvg is the standard baseline."

True but lazy — and a committee would rightly ask whether a better aggregator would have absorbed the churn damage.

The measured argument

At $K=3$, **no alternative aggregator improves on FedAvg** — and the one that theoretically should (SCAFFOLD) **fails**.

So the churn results cannot be dismissed as "you picked a weak aggregator."

And its non-IID sensitivity is a feature of the instrument

FedAvg's known fragility under heterogeneity is precisely what makes consent churn **visible** as a utility cost. A maximally robust aggregator would be the wrong measuring device — it would hide the very quantity being measured.

The central experiment: three churn regimes

The literature says “churn.” That word conflates **physically distinct** phenomena. This thesis separates them.

1. Transient

Patients withdraw, then **restore** consent later. Contributions **re-enter** the population.

Hypothesis: near-harmless.

2. Permanent

Consent withdrawn **for good**, ramping to the target rate.

Split two ways: **random** (pure shrinkage) vs. **biased** (positives leave first → distribution *shifts*).

3. Whole-silo

An **entire institution** exits — a whole slice of the distribution disappears at once.

Hypothesis: dominates.

Plus a fourth condition that is not a regime but a **control** — introduced in three slides, and it is the

How churn is injected (the mechanism)

A **schedule** is a callable $r \mapsto [K$ active-index arrays$]$ giving, for round r, the patients in each silo whose consent is currently active.

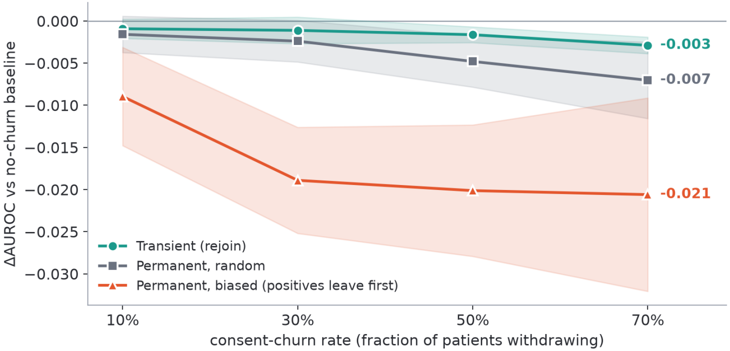
The seam

```
federated_train(..., round_silos=schedule)

for r in range(rounds):
    cur = round_silos(r)        # <- consent resolved per round
    sizes_r = [len(s) for s in cur]
    wts = sizes_r / sizes_r.sum()   # <- weights RECOMPUTED each round
    ...
```

- Silos with zero active patients contribute nothing that round.
- Aggregation weights are **recomputed every round** from active sizes — withdrawal changes the averaging, not just the data.
- **This same hook is where RQ1's gas/latency counters will attach.**

Result 1 — the rate sweep: per-patient churn is cheap



- Transient churn at **70%**: only $-\mathbf{0.003}$ AUROC. Contributions re-enter; the model barely notices.
- Permanent *random* at 70%: $-\mathbf{0.007}$. The population shrinks but does **not shift**.
- Permanent *biased* at 70%: $-\mathbf{0.021}$ — roughly $\mathbf{3}\times$ the random cost at identical headcount.

Reading the rate sweep carefully

Finding A — an honest null

Random per-patient churn is genuinely cheap on this task. Even at 70% withdrawal, the cost is under one AUROC point.

This is a real, if undramatic, finding: consortia worrying about attrition volume are worrying about the wrong thing.

Finding B — where the cost lives

The biased curve **separates immediately** and then **plateaus** near -0.021 — the damage is done early, once the positive class is depleted, not accumulated linearly with headcount.

The mechanism is **distribution shift**, not data volume.

Note the widening error band on the biased regime — with positives preferentially removed, seed-to-seed variance grows as the effective positive count shrinks. Stated, not hidden.

The isolation test: separating who from how many

The confound

A whole-silo exit removes a **distribution slice** *and* a **large headcount** at the same time. Any damage could be attributed to either. The comparison is uninterpretable as stated.

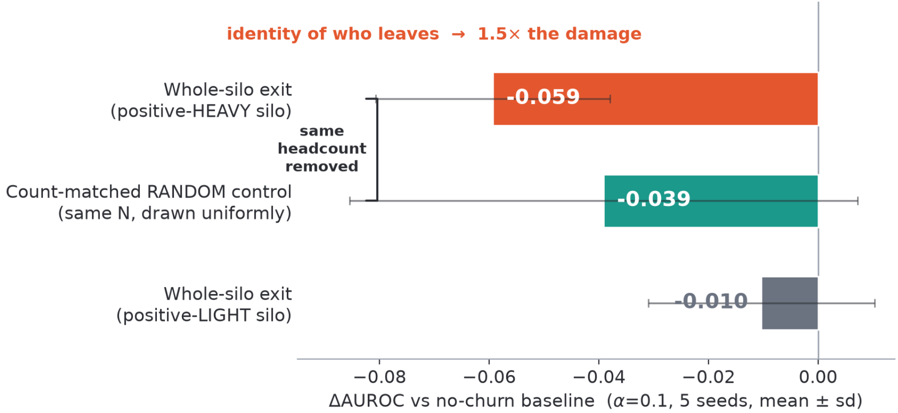
The control: `matched_random`

Remove **exactly the same number of patients** as the whole-silo exit — but draw them **uniformly at random across all silos**.

Headcount is now **held constant by construction**. Only **identity** differs.

This is the single experiment the headline claim rests on. If the two conditions came out equal, H2 would be dead.

Result 2 — the headline



Same headcount removed. **Identity carries the damage.**

H2: confirmed

Who leaves matters more than *how many* leave.

- Positive-heavy silo exit: $-\mathbf{0.059}$ AUROC.
- Same headcount, drawn at random: $-\mathbf{0.039}$.
- Positive-*light* silo exit: $-\mathbf{0.010}$ — a $\approx\mathbf{6}\times$ spread between which institution leaves.

Why this overturns the intuitive model

The natural mental model — and the one implicit in every consent-system paper — is a **headcount model**: more withdrawals, more damage. The data says the damage is **distributional**. A consortium should not be monitoring *how many* patients have opted out; it should be monitoring *whether the opt-outs have changed the case mix*.

What I will not overclaim about this result

The soft spot, stated plainly

The seed-to-seed **standard deviations are large** relative to the gap:

- heavy exit: -0.059 **(sd 0.021)**
- matched control: -0.039 **(sd 0.046)**

Five seeds on *this specific comparison* is the weakest part of the evidence.

Why the claim still stands

- The **direction** is consistent across *both* client models (next section) — an independent replication.
- The **ordering** heavy $>$ matched $>$ light reproduces in every arm.
- The $\approx$**6**$\times$ light-vs-heavy spread is far outside noise.

Committed remediation

Finer seed counts and formal significance testing (McNemar; calibration via expected calibration error) are folded into the funded work — carried over from our published benchmarking methodology.

Closing a Loophole: The Model-Capacity Check

The obvious objection — and I want to raise it myself

The doubt

Every churn number so far was measured on a **logistic-regression** client.

Maybe the damage only *looked* small because a **linear model is too blunt to feel it**. A higher-capacity model might have far more to lose under distribution shift — in which case the "cheap churn" finding is an artifact of a saturated model, not a property of consent churn.

- This objection, if correct, **invalidates the entire utility axis** — and everything RQ3 would build on top of it.
- So it was settled **before** anything else was built.

The amplification check: design

The **entire churn grid** was rerun with **two clients** on **identical** partitions and schedules:

- the original **logistic regression**;
- a **one-hidden-layer MLP** (64 ReLU units, He init, lr 0.05).

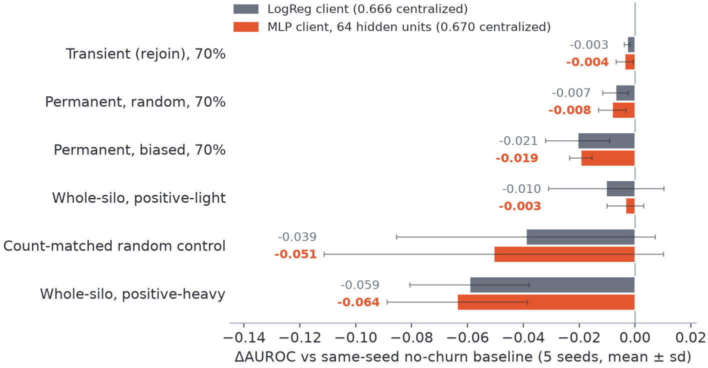
5 seeds $\times$ 3 regimes $\times$ 4 rates, plus both whole-silo exits and the matched control, $\times$ 2 clients $=$ **200 training runs**.

Two design choices that make it a fair test

1. **The MLP is genuinely stronger**, not a handicapped stand-in: it centralizes at **0.670**, *above* the linear ceiling of 0.666 and inside the published band.
2. **Aggregation is model-agnostic** — both clients are one flat parameter vector, so weighting, schedules, and harness are **unchanged**. The two arms differ *only* in the client model.

Validity check: the logreg arm of the rerun **reproduces the published churn table exactly**.

Result 3 — no amplification



Across all twelve rate-sweep cells the MLP deltas match the logreg deltas **within one standard deviation** (amplification ratios 0.2–1.6). In the **biased** regime — the one carrying the mechanism — the ratio is $\approx$**0.9**, i.e. marginally *smaller* on the MLP.

Three findings close the question

1. **No amplification — the deltas are not a linear-model artifact.** The caveat resolved in the direction that **strengthens** the result: the cost structure of consent churn is **model-independent** at this capacity.
2. **H2 survives the model swap.** On the MLP, the positive-heavy exit ($-\mathbf{0.064}$) still dominates the count-matched control ($-\mathbf{0.051}$) at identical headcount, and dwarfs the positive-light exit ($-\mathbf{0.003}$) — an $\approx \mathbf{18}\times$ who-leaves gap. "Who leaves matters more" is **not a property of logistic regression**.
3. **Why nothing amplifies: the task ceiling, not the model class, bounds the deltas.** The MLP lifts the averageable ceiling only $0.666 \to 0.670$ against a tree ceiling of 0.677. On a task **at its signal ceiling** there is little headroom for *any* model to lose dramatically more under shift.

One secondary observation worth reporting

Higher capacity is slightly more fragile at baseline

The MLP's **no-churn** baseline is marginally worse under severe skew:

α=0.1: MLP **0.629** vs. logreg **0.640**

Higher capacity overfits skewed silos marginally faster.

- This cuts *against* the "use a bigger model" reflex: at K=3 under severe heterogeneity, capacity is a mild liability **before churn is introduced at all**.
- It also explains part of why the MLP's churn deltas do not amplify — it starts from a lower, noisier baseline.

The utility axis is closed

Summary of RQ2

On a real clinical task at its signal ceiling:

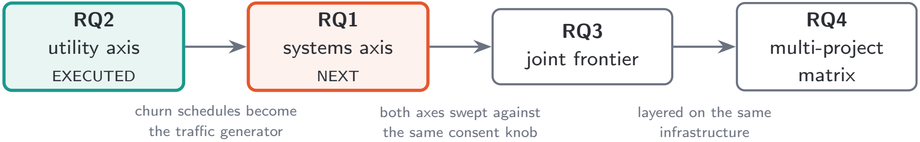
- **per-patient consent churn stays cheap** (-0.003 to -0.007 even at 70% withdrawal);
- **distribution-shifting churn is 3–6$\times$ more expensive**;
- and this holds **regardless of client capacity**.

RQ2 executed and closed, H2 confirmed, model-robustnes

The remaining external-validity question is the **Dirichlet partitioning**, not the model class — addre

Proposed Work

The critical path



The key economy

RQ1 does **not** need new experimental design. The **already-built churn schedules** generate the on-chain transaction traffic. The same knob that drives the utility axis drives the systems axis — which is what makes a *joint* frontier possible at all.

RQ1 — what gets instrumented

Every federated round is instrumented to decompose its cost into **compute** and **governance**.

Governance covers

- **Consent-set resolution** — querying the ledger for who is active this round
- **Grant/revoke transactions** — gas per operation
- **Encrypted-record retrieval** — IPFS fetch latency

Measured per round

- wall-clock latency (distribution, not mean)
- gas / monetary cost
- bandwidth
- failure rate under congestion

Attachment point

The `round_silos` hook shown earlier — the same seam the churn schedules already use. **No architectural change is required.**

RQ1 — the sweep

Swept parameter	Range and rationale
Consent-churn rate	10–70%, reusing the executed schedules — ties the systems axis to the utility axis on *one shared knob*
Consent-refresh cadence	every round $\rightarrow$ every n rounds — the amortization dimension that H1 predicts will become necessary
Federation size K	small $\rightarrow$ hospital-scale, driven by **Synthea** populations
Chain backend	**local** (Ganache), **permissioned**, **public** (Sepolia) — spanning the 0.05 s to 108 s range already measured

Reporting commitment

Distributions, not means. Public-chain latency is heavy-tailed and we have already observed outright transaction failures under congestion — a mean would erase exactly the behavior that determines feasibility.

RQ1 — what a positive result looks like

The H1 objective

Locate the **infeasibility threshold**: the churn rate beyond which per-round on-chain consent enforcement **cannot keep up**, and consent must be **amortized across rounds**.

- The threshold is expected to be **backend-dependent** — and the published measurements already suggest the spread is enormous (0.05 s local vs. up to 108 s public per operation).
- Amortization is not free: it introduces **consent staleness**. Training proceeds on a consent set that is knowingly out of date.
- **The deliverable is the amortization–staleness tradeoff curve** — how much staleness must be accepted to stay feasible, per backend.

An ethically loaded quantity — and exactly what the architectural literature never surfaces.

RQ3 — assembling the joint frontier

Both axes are swept against the **same** consent knobs, then plotted against each other.

- x: governance cost per round (from RQ1)
- y: model utility (from RQ2)
- parameterized by: churn rate, **refresh cadence**, K, backend

H3 — the knee

Some refresh cadences should recover **most** of the utility of per-round enforcement at **a fraction** of the cost.

Why RQ2's result sharpens this

The utility axis already told us **where utility is sensitive**: to **distribution shift**, not headcount.

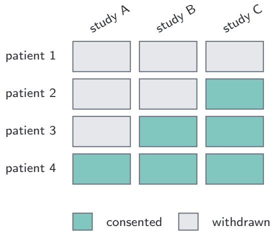
So the frontier's shape should be governed by how quickly consent staleness lets the **case mix** drift — not by how many revocations accumulate.

That is a **sharper, falsifiable prediction** than we could have made before RQ2 was executed.

RQ4 — multi-project consent

A patient is rarely in *one* study. The consent layer must govern a **patient** $\times$ **study matrix**.

- Implemented at the **contract/ledger level**.
- Measure the **marginal governance cost** per additional study.



H4

Overhead is **bounded and sub-linear**, because consent state is **shared infrastructure**, not per-study duplication.

Demonstrates generality **without building a second end-to-end system** — the marginal cost is the evidence.

Timeline, Risks, and Limitations

Timeline

Period	Milestone
2024 *(done)*	Component papers published: Spark benchmarking, blockchain/IPFS consent layer, supply-chain regime shift.
Spring 2026 *(done)*	Thesis framing finalized (two-axis cost of autonomy); primary dataset locked (Diabetes 130); Synthea confined to the systems axis.
Summer 2026 *(done)*	Utility axis executed: centralized ceiling, five-method federated comparison, three-regime churn study with the count-matched isolation test (H2 confirmed), and the MLP amplification check.
Fall 2026	**Systems-cost axis (RQ1)**: gas/latency/bandwidth instrumentation on the per-round hook across local, permissioned, and public backends; Synthea scaling to locate the H1 threshold.
Winter 2026–Spring 2027	**Joint frontier (RQ3)**: both axes swept against the shared churn knob; locate the amortization knee. **Multi-project matrix (RQ4)**.
Spring–Summer 2027	Robustness replication on real hospital silos (eICU); dissertation writing and defense.

Risks and mitigations

Risk	Mitigation
Public-testnet volatility makes RQ1 measurements irreproducible	Report distributions across time-of-day (methodology already validated); anchor with the deterministic local backend; treat congestion failures as *data*, not noise
Dirichlet partitions are not real hospital boundaries	Real-silo replication on **eICU** is scheduled, not optional; the partitioning is named as the remaining external-validity threat
Utility deltas are moderate in absolute terms	Already de-risked: the model-class explanation was tested and **eliminated**; the deltas are bounded by the *task ceiling*, which is a property of the domain
Synthea populations do not reflect real consent behavior	Synthea is confined to the systems axis, where only **volume and payload size** matter and signal realism is irrelevant
Frontier turns out to be linear (H3 false)	A linear frontier is still a publishable measurement — it would mean amortization buys nothing, which contradicts current deployment practice

Limitations, stated plainly

- The federation is **simulated**, not a live multi-hospital deployment.
- Silos are **Dirichlet partitions of one cohort**, not real hospital boundaries. This is now the **primary** external-validity question — the model-class alternative having been ruled out.
- Measured utility magnitudes are **moderated** by the large cohort and by AUROC's rank-robustness.
- Synthetic data is confined to the systems axis — by design, but it does mean the systems axis is not driven by real consent behavior.
- The Year-1 system claims **consent governance and data autonomy**, **not formal privacy**.

Each of these is either scheduled for remediation (eICU, DP thrust) or is an accepted, named boundary of the contribution.

Future work beyond the proposal

1. **Privacy that earns the word.** Compose DP-SGD and/or secure aggregation, adding a **third axis** (noise $\rightarrow$ accuracy loss) to the same frontier.
2. **Federated unlearning.** The *biased permanent withdrawal* regime is the natural setting: when a patient revokes, can their contribution be **removed** from an already-trained model rather than merely excluded going forward?
3. **Real hospital silos.** eICU replication, replacing Dirichlet partitions with genuine institutional boundaries.
4. **Image scale.** Cervical images (SipakMed), where the published Spark/CNN pipeline is **justified** by the compute-placement rule.

Summary

What is done [EXECUTED]

- Centralized ceiling, local floor, federated ceiling — all measured
- Five aggregation methods compared; FedAvg empirically justified
- Three churn regimes separated and swept
- **H2 confirmed** by count-matched isolation test
- **Model-independence verified** (200 runs)

What is proposed [PLANNED]

- RQ1: on-chain governance instrumentation across three backends
- H1 infeasibility threshold and the amortization–staleness tradeoff
- RQ3: the joint cost–utility frontier and its knee
- RQ4: multi-project consent matrix, marginal cost per study

Patient autonomy is not a feature to be built — it is a **cost to be measured**.
One axis is measured. This proposal builds the other, and joins them.

Thank you.

Questions?

Backup Slides

Backup — full churn results table

Condition	LogReg client	MLP client
Transient (rejoin), 70% churn	−0.003 (0.001)	−0.004 (0.003)
Permanent, random, 70%	−0.007 (0.005)	−0.008 (0.005)
Permanent, biased, 70%	−0.021 (0.011)	−0.020 (0.004)
Whole-silo exit, positive-light (α=0.1)	−0.010 (0.021)	−0.004 (0.007)
Whole-silo exit, positive-heavy (α=0.1)	−0.059 (0.021)	−0.064 (0.025)
Count-matched random control (α=0.1)	−0.039 (0.046)	−0.051 (0.061)
No-churn baselines		
α=0.5	0.6587	0.6575
α=0.1	0.6404	0.6290

ΔAUROC vs. the **same-seed** no-churn baseline under identical partitions and consent schedules; mean (sd) over five seeds.

Backup — hyperparameters

Federated setup

- Rounds: **40**; local epochs: **1**
- Batch size: **256**; L_2: **1e-4**
- Class weighting: **on** (global, from train split)
- $K = 3$ silos; seeds **0–4**
- Learning rate: **0.1** (logreg), **0.05** (MLP)

Method-specific

- FedProx: $\mu = 0.1$
- FedAvgM: server momentum 0.9
- FedAdam: server lr 0.05, β_1=0.9, β_2=0.99, ϵ=1e-3
- SCAFFOLD: Option II control-variate update
- Biased schedule: withdrawal bias $= 8.0\times$ for positives

MLP: 1 hidden layer, 64 ReLU units, He initialization, flat parameter vector of size $n_f \cdot H + 2H + 1$.

Backup — the four churn schedules

Schedule	Mechanism
`transient`	Each round independently deactivates ≈ rate of every silo; patients **can return**. Fresh RNG per round.
`permanent`	Cumulative, irreversible withdrawal ramping to rate by the final round. Withdrawal order **random within each silo** ⇒ shrinkage without shift.
`biased_permanent`	Same ramp, but positive-class patients withdraw preferentially ($8\times$), via a Gumbel-top-k / exponential-race weighted ordering ⇒ **shifts the class distribution**.
`whole_silo`	Named silos hold **no** active patients from `depart_round` on.
`matched_random`	Removes **exactly** `n_remove` patients drawn uniformly across *all* silos — the count-matched control.

Backup — why SCAFFOLD collapses at α=0.1

- SCAFFOLD corrects client drift using **control variates** c_i estimated from each client's local updates.
- Those estimates are only reliable when clients have **enough data** and **enough participation** to average out noise.
- At α=0.1, K=3: one silo has **529 patients at 2.3% positive**, another **4,273 at 82% positive**.
- Control variates estimated from such extreme, small silos **inject noise** rather than correcting drift — a documented small-K failure mode.

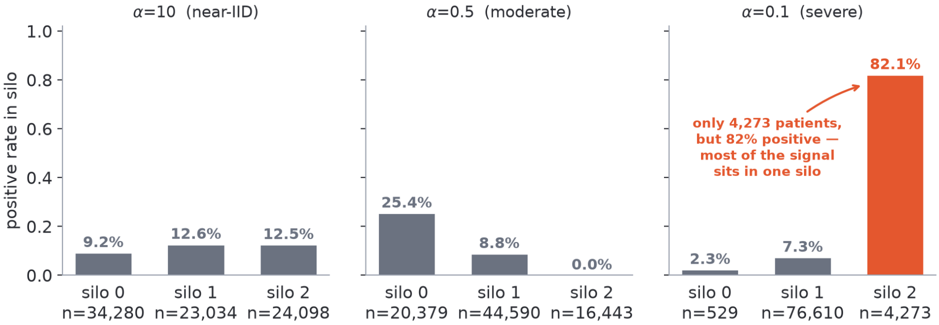
Why this is reported rather than dropped

It is a **diagnostic**, not a nuisance: it confirms the α=0.1 regime is genuinely pathological, which is what makes it the right stress condition for the whole-silo exit test.

Backup — full five-method federated results

	AUROC			
Method	α=10	α=0.5	α=0.1	PR-AUC (α=0.1)
FedAvg	0.6689	0.6677	0.6662	0.2110
FedProx	0.6682	0.6669	0.6664	0.2106
FedAvgM	0.6673	0.6669	0.6666	0.2099
FedAdam	0.6671	0.6655	0.6665	0.2112
SCAFFOLD	0.6700	0.6689	**0.6246**	**0.1758**
Reference anchors				
Centralized averageable ceiling (logreg)	0.6662			
Best centralized overall (Random Forest)	0.6770			
Local-only floor (α=0.5, mean)	0.6157 (min 0.5159)			

Backup — Dirichlet partition composition



At α=0.5, silo 2 receives **zero** positive cases — an extreme but legitimate draw that the class-weighted trainer must handle (its contribution to the positive gradient is nil, and aggregation weights are recomputed from active sizes each round).

Backup — blockchain cost detail

Ganache (local, single node)	Time (s)	Cost (ETH)
Add permission	0.052096	0.00099022
Revoke permission	0.070517	0.00119175
Sepolia (public testnet)		
Add, fastest	8.84	—
Revoke, fastest	12.01	—
Add, peak (12 PM)	55.09	up to 0.0029
Revoke, peak (1–2 AM)	107.99	up to 0.019805

Congestion between 6–9 AM CT produced outright transaction **failures** under the configured wallet settings. IPFS retrieval typically under 0.3 s. Revoke is consistently costlier than add on both backends — a detail that matters, since **revocation is the operation patient autonomy generates**.

Backup — reproducibility

- All experiments run from a single conda environment (`thesis`, Python 3.11).
- `01_baseline_fullscale/` — preprocessing, multi-model centralized benchmark, five FL aggregation methods.
- `02_consent_churn/` — churn schedules, the executed three-regime study, and the MLP amplification runner.
- Saved result files (`*.json`) hold every per-run metric; the figures in this talk are **regenerated from those files**, not transcribed.
- Dataset auto-downloads from the UCI repository on first run; raw data is kept out of version control.

Release plan

Testbed, contracts, churn-schedule generators, and Synthea-derived populations released open-source (C5), so the cost–utility measurements can be reproduced and extended.